UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PROGRAMMING I
COURSE CODE	:	CSC402
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC402 mock paper (November 2026). Generated answers show working; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 10 x 1 mark; Part B = 5 questions (60 marks); Part C = 2 x 15 marks; total = 100 marks.

4. Questions: 17. Slots: 22. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 21 unique CSC402 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A — ANSWER KEY

Q	Ans	Question	Solution
A1	D	Which statement immediately exits a loop?	break terminates the nearest loop or switch.
A2	D	What is the index of the first element of a C++ array?	C++ arrays are zero-indexed.
A3	C	How many times does a while loop execute when its condition is false initially?	while is a pretest loop; a false condition skips the body.
A4	A	Which structure chooses between two alternative paths?	if...else performs two-way selection.
A5	D	Which declaration creates a floating-point variable named price?	float is used for real numbers.
A6	D	Which values can a bool variable hold?	A bool holds true or false.
A7	A	Which term describes a finite sequence of precise steps that solves a problem?	An algorithm is a finite, ordered set of unambiguous steps.
A8	C	Which item is NOT a programming paradigm?	Debugging is an activity, not a programming paradigm.
A9	D	Which identifier is valid in C++?	An identifier may contain letters, digits and underscores, but cannot start with a digit or be a keyword.
A10	B	Which input pattern stops when the user enters -1?	A sentinel value ends the loop.

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PART B — WORKED ANSWERS

QUESTION 1

Topic: debugging testing

a) Define compiler and interpreter.

A compiler translates the whole source program before execution; an interpreter translates and executes one statement at a time.

Marking: Award 1 mark for each correct definition.

(3 marks)

b) State THREE differences between a compiler and an interpreter.

Compiler: whole program, produces object code, reports errors after compilation. Interpreter: statement by statement, no object file, stops at the first error; generally slower execution.

Marking: Award 1 mark for each valid difference.

(2 marks)

QUESTION 2

Topic: fundamentals

a) Draw a flowchart to find the largest of three values.

Start -> read a, b, c -> largest = a -> if b > largest then largest = b -> if c > largest then largest = c -> display largest -> end.

Marking: Award marks for input, initial assumption, two comparisons, output and flow.

(6 marks)

b) Write pseudocode and a complete C++ program for the flowchart.

Use if statements to update largest, then cout the result.

Marking: Award marks for correct comparisons, variable updates and output.

(7 marks)

QUESTION 3

Topic: data types

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

int age; double price; const double DISCOUNT = 0.10; char grade;

Marking: Award marks for correct types, names and const initialisation.

(5 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

cin >> age >> price; price = price - price * DISCOUNT; cout << price;

Marking: Award marks for input, discount expression and output.

(6 marks)

QUESTION 4

Topic: input output

a) Trace the program and state the output.

30

Marking: The sum of 2+4+6+8+10 is 30.

(10 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

int i=0,s=0; do { if(a[i]>5) s+=a[i]; i++; } while(i<5); cout<<s;

Marking: Award marks for do...while, condition, filter and output.

(5 marks)

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QUESTION 5

Topic: arrays

a) Declare an integer array of ten marks and read the marks from the user.

int marks[10]; for(int i=0;i<10;i++) cin >> marks[i];

Marking: Award marks for array declaration, loop and input.

(6 marks)

b) Write a program segment to calculate and display the average, highest mark and number of marks above the average.

Compute total and highest in one loop; average = double(total)/10; then second loop counts marks > average.

Marking: Award marks for total, average, highest, second loop and output.

(10 marks)

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PART C — WORKED ANSWERS

QUESTION 1

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

```
#include <iostream>
using namespace std;
int main() {
    int marks[10], total = 0, highest, highestIndex = 0;
    int aboveAverage = 0;
    for (int i = 0; i < 10; i++) {
        cin >> marks[i];
        total += marks[i];
        if (i == 0 || marks[i] > highest) {
            highest = marks[i];
            highestIndex = i;
        }
    }
    double average = total / 10.0;
    for (int i = 0; i < 10; i++) {
        if (marks[i] > average) aboveAverage++;
    }
    cout << "Average: " << average << endl;
    cout << "Highest: " << highest
         << " at index " << highestIndex << endl;
    cout << "Above average: " << aboveAverage << endl;
    return 0;
}
```

Marking: Award 3 marks for input, 4 for total/average, 3 for highest/index, 3 for above-average count and 2 for output.

(15 marks)

QUESTION 2

A catering company charges RM12.50 per meal. Orders of 50-99 meals cost RM11.50 each, 100-199 cost RM10.50 each, and 200 or more cost RM9.50 each.

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

```
#include <iostream>
#include <iomanip>
#include <string>
using namespace std;
int main() {
    string name;
    int quantity, grandQuantity = 0;
    double unitPrice, total, grandTotal = 0;
    char again;
    do {
        cout << "Customer name: ";
        getline(cin >> ws, name);
        cout << "Quantity: ";
        cin >> quantity;
        if (quantity < 0) {
            cout << "Invalid quantity" << endl;
        } else {
            if (quantity >= 200) unitPrice = 9.50;
            else if (quantity >= 100) unitPrice = 10.50;
            else if (quantity >= 50) unitPrice = 11.50;
            else unitPrice = 12.50;
            total = quantity * unitPrice;
            grandTotal += total;
            grandQuantity += quantity;
            cout << fixed << setprecision(2);
```

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```
            cout << "Receipt for " << name << ": "
                 << quantity << " x RM" << unitPrice
                 << " = RM" << total << endl;
        }
        cout << "Another customer? (y/n): ";
        cin >> again;
    } while (again == 'y' || again == 'Y');
    cout << "Total quantity: " << grandQuantity << endl;
    cout << "Total sales: RM" << fixed << setprecision(2)
         << grandTotal << endl;
    return 0;
}
```

Marking: Award 3 marks for input/validation, 6 for price selection and total, 3 for repetition, 2 for final totals and 1 for output clarity.

(15 marks)

END OF MARKING SCHEME

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